

Dhaval Patel

☎ +1 (878)-245-7847 ✉ dmpatel2702@gmail.com in in/dhavalp210/ 🌐 dhaval-portfolio-zeta.vercel.app/ 📄 dhaval270

EDUCATION

University of Massachusetts Amherst – Amherst, MA

September 2025 – May 2027

Master of Science in Computer Science

GPA: 3.76/4.0

Relevant Coursework - Advance Natural Language Processing, Machine Learning, Neural Networks, Software Engineering

Charotar University of Science and Technology – Anand, India

October 2021 – May 2025

Bachelor of Technology in Computer Engineering

GPA: 9.0/10

Relevant Coursework - Data Structures, Database Management System, Operating System, AI, Image Processing

PROFESSIONAL EXPERIENCE

Cilans System

July 2024 - June 2025

AI/ML Engineer

Ahmedabad, India

- Architected and deployed scalable backend pipelines for the **GranthAI** document processing platform, integrating APIs with OpenAI and Gemini to handle high-volume document ingestion and processing.
- Designed and integrated Azure Cosmos DB with scalable indexing and retrieval workflows, supporting multilingual document search at scale with sub-second query response times.
- Reduced end-to-end system latency by 47% by optimizing API call orchestration, implementing efficient caching strategies, and refactoring monolithic workflows into modular components.
- Built RESTful APIs and microservices using FastAPI for document parsing, summarization, and Q&A functionality, ensuring high availability, modular architecture, and seamless downstream integration with frontend services.
- Collaborated in an Agile environment with cross-functional teams, owning feature delivery from design to deployment using Git workflows, peer code reviews, Docker-based containerization, and CI/CD pipelines for reliable, reproducible releases.
- Engineered proof-of-concept comparing LLM-based OCR with Python OCR libraries, demonstrating 15-20% improvement in accuracy and validating feasibility for production migration.
- Partnered with backend engineers and client-facing PM to standardize invoice parsing across 8 client formats, increasing extraction accuracy by 21%.

Charotar University of Science and Technology (CHARUSAT)

May 2024 - July 2024

Machine Learning Research Intern

Anand, India

- Streamlined construction resource estimation accuracy by 24% by extracting dimensional information from floor plan images using OCR and optimizing material requirement calculations for greater precision and efficiency.
- Co-authored 3 papers with two research labs on applied deep learning, presenting findings on BERT-based cognitive skill evaluation, YOLOv8-driven real-time activity detection, and novel approaches to improving model accuracy, efficiency, and real-world applicability across educational and sports analytics domains.

TECHNICAL SKILLS

Programming Languages: Python, R, C++, HTML, CSS, SQL, JavaScript, Node.js

Data Science and Machine Learning: Deep Learning, NLP, LLMs, RAG, Agentic Systems (LangGraph), Prompt Engineering, Model Deployment, Time Series Analysis

Tools & Technologies: Git, Docker, Amazon Web Services (EC2, S3), Microsoft Azure, Claude Code

Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, LangChain, FastAPI, OpenCV, Pandas, NumPy

PROJECTS

DistroTrack — Distributor Management Platform (Next.js 14, Supabase, TypeScript, Tailwind CSS) [Link]

- Digitalized manual notebook workflows across sales, inventory, sales, membership tracking, and dashboard analytics with low-stock alerts and volume point tracking.
- Generated period reports for profit, revenue, and inventory with printable summaries, giving distributors accurate figures for business decisions in place of a slow, error-prone paper process.

LedgerLens — Agentic personal finance analyst (Python, LangGraph, SQLite, Groq API) [Link]

- Built a multi-node LangGraph agent that decomposes natural-language finance questions, routes across SQL, semantic and statistical tools, and self-corrects failed queries, reaching 80.4% execution accuracy across 46 benchmark queries with 100% first-attempt SQL validity, while correctly refusing all 7 deliberately unanswerable ones.
- Built a deterministic verification node rejecting answers whose figures don't trace to a retrieved row: 100% catch rate on 126 injected wrong numbers, 0 false rejections, surfacing hallucinated financial totals instead of passing them silently.

Robust Anomaly Detection under Real-World Image Corruptions (PyTorch, PaDiM, PatchCore)[Link]

- Investigated robustness of industrial anomaly detection models (PaDiM, PatchCore) built on Wide-ResNet-50 against real-world image corruptions such as Gaussian noise, blur, brightness, JPEG across 5 severity levels on the MVTec AD dataset, revealing critical performance gaps where image-level ROC-AUC collapsed to 0.36 below random under brightness corruption for texture categories.
- Designed a corruption-aware data augmentation pipeline that improved model robustness by up to 30% by training memory banks and Gaussian distributions on mildly corrupted normal samples in PyTorch.

PUBLICATIONS

• **Impact of BERT on Evaluating the Quality of Question Paper Using Bloom's Taxonomy** - ICT4SD-2024
DOI: 10.1007/978-981-97-8526-1_6

• **Real-Time Sports Analysis: Integrating YOLOv8 with Enhanced Feature Pyramid Networks for Volleyball and Cricket Activity Detection** - ADCIS-2024 DOI: 10.1007/978-981-96-4536-7_9